

Introduction

Drug-Drug Interactions a reaction between two (or more) drugs can be due to pharmacokinetics or pharmacodynamics properties of medication. Unrecognized drug-drug interactions can lead to morbidity and mortality and withdrawal of approved drugs from the market. Understanding how to safely manage a Drug-Drug Interaction (DDI) may allow healthcare practitioners to minimize associated risk and patient noncompliance.

Scope

This document is applicable to Pharmacy professionals and other linked health care workers dealing with clinical area in royal hospital.

Purpose

1. To outline general guidance for dealing with Drug-Drug Interactions in the pharmacy.
2. To minimize Drug-Drug Interactions related to the use of medications and improve the overall health care of the patients.

Definitions

Drug-Drug interactions: A pharmacokinetics or pharmacodynamics influence of drugs on each other, which can result, beside desired effects, in reduced effectiveness or increased toxicity.

Pharmacokinetics interaction: is when one medication alters the absorption, distribution, metabolism and/or elimination of other medication.

Pharmacodynamics interaction: is when one medication changes the Pharmacological response of another medication in an additive, synergistic or antagonistic way.

Policy

1. All pharmacists and healthcare providers should take the responsibility for the prevention and resolution of Drug-Drug Interactions (DDIs).
 - Physician must justify and review each drug regularly, screen for DDIs with each drug's addition or deletion,

- Nurses must assess and monitor drug administration and document any adverse drug reaction or change in the patient status and report for the same.
2. Clinical Pharmacist/Pharmacist must review each medication order for DDIs, assist in Drug selection or removal and monitor for any adverse drug reaction or interaction.
 3. DDIs determined by the Clinical Pharmacist/ Pharmacist must categorized into three categories:
 - Contraindicated combination: The drug combination is clearly contraindicated in all cases, should be documented in the file and not be dispensed to the patient.
 - Major interaction: Action is required to reduce the risk of severe Adverse interaction/reaction.
 - Moderate interaction: Assess risk to patient and take action as needed.
 4. For Drug-Drug interactions, the approach must be the following:
 - Take the medication history.
 - Keep in mind high risk patients.
 - Evaluate therapeutic alternatives:
 - Dose spacing
 - Decreasing the dose
 - Discontinue the drug
 - Add another drug to counter the interaction
 5. Reporting of DDI must be done in Adverse Drug Reaction Form which is available in the local site of the hospital.
 6. Additional consideration is required when one medication with narrow therapeutic index is prescribed or involved (e.g. warfarin).

Procedure

1. Check for **interactions** including **drug–drug**, **drug–patient**, **drug–disease** and **drug–nutrient** interactions and identify **mechanism** of the interaction & consider clinical **significance**. Decide on an appropriate course of **action**.
2. Consider **interactions** with **laboratory tests** and **environmental** factors, e.g. smoking, alcohol consumption, motor vehicle driving.
3. Clinical pharmacist/pharmacist may assist in the detection and interpretation of DDI data and identifying an alternative treatment if available within the medications in the hospital formulary through daily prescription review and medicine reconciliation process.

4. Several electronic databases such as Mobile Apps Micromedex, uptodate, lexicomp, Medscape, and Drug.com are useful.

References

Title of book/journal/article/website	Author	Year of publication	Page
Pharmacokinetics and pharmacodynamics of drug interactions involving rifampicin, rifabutin and antimalarial drugs.	Marta Sousa, Anton Pozniak, Marta Boffito	2008	872-878
Standards of Practice for Clinical Pharmacy Services. Journal of Pharmacy	George Taylor	2012	43
Common drug interactions leading to adverse drug events in the intensive care unit: management and pharmacokinetic considerations.	Papadopoulos J1, Smithburger PL	2010	
Drug Risks and Benefits	Chris J.vanBoxtel, Budiono Santoso,	2008	
Drug Interaction Pairs Associated with an increased likelihood of Hospitalization: A New Look at the Evidence.	I.Ralph Edwards Roger Cheng	2012	